



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

UNIT TEST

19-09-2021

JEE(Main + Advanced) : Nurture Course (PHASE : TEPS, I, I(B), II(A), II & LIVE-I, I(B), II(A))

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	B	B	D	A	C	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	A	D	B	D	D	D	B	C	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	4.5	3.00	800.00	15.00	3.00	3	16	3.20	8.00	1.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	C	D	B	A	C	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	D	B	C	D	B	D	D	A	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	3.00	4.00	32.00	7.50	1.60	6.00	4.00	6.00	2.00	6.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	A	B	C	B	D	B	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	A	D	A	A	B	B	D	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1.00	3.00	-4.00	9.00	5.00	2.00	16.50	3.00	3.00	5.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (B)**

$$\Sigma I = mr_1^2 + mr_2^2 + \dots + mr_n^2 = m\ell^2 + m(2\ell)^2 + \dots + m(n\ell)^2$$

$$\Rightarrow m\ell^2[1^2 + 2^2 + \dots + n^2] = \frac{m\ell^2}{6}n(n+1)(2n+1)$$

2. **Ans (C)**

$$I = 4mH^2 + 4m(H^2 + R^2)$$

3. **Ans (C)**

$$I = 2(M \times 0) + 4(M \times a^2) + 2[M \times (\sqrt{2}a)^2]$$

$$= 8Ma^2$$

4. **Ans (B)**

$$F = 600 - 2 \times 10^5 t$$

$$F = 0, t = \frac{600}{20 \times 10^5} = 3 \times 10^{-3} \text{ sec}$$

$$I = \int_0^t F \cdot dt = \int_0^t (600 - 2 \times 10^5 t) dt$$

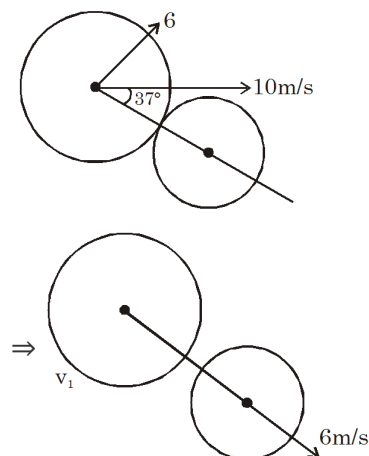
$$I = [600t - 10^5 \cdot t^2]_0^t$$

$$I = 600 \times 3 \times 10^{-3} - 10^5 \times 9 \times 10^{-6}$$

$$1.8 - 0.9 = 0.9 \text{ N-S}$$

5. Ans (B)

By momentum conservation along line of impact,



$$m \times 8 = m \times v_1 + m \times 6$$

$$\Rightarrow v_1 = 2 \text{ m/s}$$

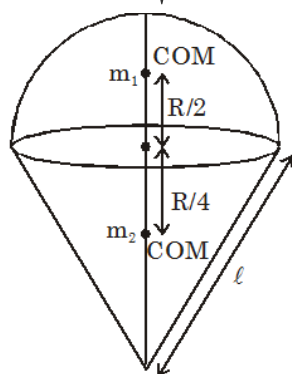
$$\therefore e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{6 - 2}{8} = \frac{1}{2}$$

6. Ans (D)

$$\text{Mass of shell } (m_1) = \sigma(2\pi R^2)$$

$$\text{Mass of cone } (m_2) = \sigma\pi R \ell = \sigma\pi R \sqrt{R^2 + \left(\frac{3R}{4}\right)^2}$$

$$m_2 = \sigma\pi R^2 \sqrt{\frac{16+9}{16}} \quad m_2 = \frac{5}{4}\pi\sigma R^2$$



x is distance of $\text{COM}_{(\text{System})}$ from m_1

$$x = \frac{m_2 \left(\frac{3R}{4}\right)}{(m_1 + m_2)} = \frac{\left(\frac{5}{4}\pi\sigma R^2\right) \left(\frac{3R}{4}\right)}{\left(\frac{5}{4}\pi\sigma R^2 + 2\pi\sigma R^2\right)}$$

$$x = \frac{\left(\frac{5}{4} \times \frac{3R}{4}\right)}{\left(\frac{5}{4} + 2\right)} = \frac{\frac{15}{16}R}{\left(\frac{5+8}{4}\right)} = \frac{15}{16} \times \frac{4R}{13} = \frac{15}{52}R$$

Distance of $\text{COM}_{(\text{System})}$ from vertex of cone

$$= \frac{3R}{4} + \left(\frac{R}{2} - \frac{15R}{52}\right)$$

$$= \frac{3R}{4} + \frac{(26R - 15R)}{52}$$

$$= \frac{3R}{4} + \frac{11R}{52} = \frac{(39 + 11)}{52} = \frac{50}{52}R = \frac{25}{26}R$$

7. Ans (A)

$$I = m_1 r_1^2 + m_2 r_2^2$$

$$= 5(2)^2 + 3(2)^2$$

$$= 8(2)^2$$

$$= 32 \text{ kg-m}^2$$

8. Ans (C)

At the highest point and lowest point the component of normal reaction between M and m in the horizontal is zero. But in between these points, normal reaction in the horizontal direction was non-zero. That means normal reaction in horizontal direction first increases then decreases and so the acceleration of mass 'M'.

9. Ans (D)

at $t = 1 \text{ sec}$

10
↑
● → 20 m/s
before explosion

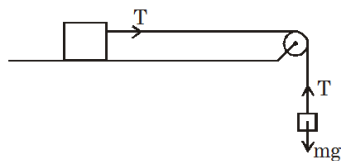
at $t = 1 \text{ sec}$

20
↑
● → 40
after explosion

$$H_{\text{max}} = \left[20 \times 1 - \frac{1}{2} \times 10 \times 1^2\right] + \left[\frac{(20)^2}{2 \times 10}\right]$$

$$= 15 + 20 = 35 \text{ m}$$

10. **Ans (B)**



$$2mg - T = 2ma$$

$$T = ma$$

$$a = \frac{2g}{3}$$

$$\vec{a}_{\text{com}} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2}{m_1 + m_2} = \frac{m \frac{2g}{3} \hat{i} - 2m \frac{2g}{3} \hat{j}}{3m}$$

$$= \frac{2g}{9} \hat{i} - \frac{4g}{9} \hat{j}$$

11. **Ans (C)**

The forces are equal. Newton's 3rd Law:
Action/Reaction.

12. **Ans (A)**

$$\frac{1}{2}mv^2 = \mu mgx + \frac{1}{2}kx^2 \quad \dots (i)$$

$$2\mu mg = kx \quad \dots (ii)$$

From (i) and (ii)

$$v = \mu g \sqrt{\frac{8m}{k}} = 2\mu g \sqrt{\frac{2m}{k}}$$

13. **Ans (D)**

$$\frac{1}{2}kx^2 = \frac{1}{2} \frac{m \times 3m}{4m} (v - 0)^2$$

14. **Ans (B)**

Given $u_2 = 0$

$$\text{we know } v_1 = \frac{(m_1 - em_2) u_1}{m_1 + m_2}$$

Since $m_1 > em_2$, hence v_1 is +ve, so ball will move in the same direction.

16. **Ans (D)**

$$T_A \cos \theta = mg, T_B = mg \cos \theta$$

$$\frac{T_B}{T_A} = 10s^2 \theta = \frac{1}{4}$$

17. **Ans (D)**

$$\text{Power} = Fv = m \frac{dv}{dt} (v) \propto v$$

$$\Rightarrow mv \frac{dv}{dt} = k_0 v$$

where $k_0 = \text{constant}$

$$\Rightarrow mv \frac{dv}{dt} = k_0$$

$$\Rightarrow mv \frac{dv}{dx} = k_0$$

$$\Rightarrow v dv \frac{k_0}{m} dx$$

Integrating both sides

$$\int_0^v v dv = \frac{k_0}{m} \int_0^x dx$$

$$\frac{v^2}{2} = \frac{k_0}{m} x$$

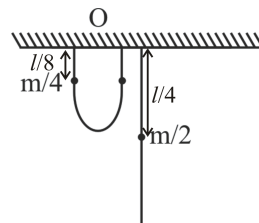
Hence, displacement is proportional to square of instantaneous velocity. Hence (D) is correct

18. **Ans (B)**

$$T = \frac{mv^2}{\ell}$$

19. **Ans (C)**

after taking point P to O



COM of system from O

$$y_{\text{cm}} = \frac{m/4 \times l/8 + m/4 \times l/8 + m/2 \times l/4}{m}$$

$$= \frac{3\ell}{16}$$

$$\Delta \text{cm} = \frac{\ell}{2} - \frac{3\ell}{16}$$

$$= \frac{5\ell}{16}$$

$$\text{work done by gravity} = \frac{5mg\ell}{16}$$

20. **Ans (A)**

Speed after t time $v = a_t$

$$a_c = a_t$$

$$\frac{v^2}{R} = 2.$$

$$\frac{a^2 t^2}{2} = 2 \quad \frac{4t^2}{2}$$

$$t = 1 \text{ sec}$$

SECTION-II

2. Ans (3.00)

Velocity of B after collision will be

$$V = \frac{3}{2} \frac{2mv_0}{3m} = v_0$$

Tension just after the collision is

$$T = \frac{mv_0^2}{\ell} + mg = 40$$

$$\Rightarrow v_0^2 = 30 \times 0.3$$

$$\Rightarrow v_0 = 3 \text{ m/s}$$

3. Ans (800.00)

By conservation of momentum

$$-200 \times v_1 + 2 \times v_2 = 0$$

$$100 v_1 = v_2 \quad \dots (1)$$

$$v_1 = v_2 / 100$$

$$\left(\frac{v_2}{100} \right) \cdot t = 8$$

$$t = \frac{800}{v_2}$$

$$v_2 \times t = x$$

$$x = v_2 \times \frac{800}{v_2}$$

$$x = 800 \text{ m}$$

4. Ans (15.00)

Under area of $F - t = \Delta P$

so, P will be maximum at $t = 15 \text{ sec}$.

5. Ans (3.00)

I collision : between B and C.

II collision : between C and D.

(After that $v_C = 0$)

III collision : between B and A.

(After that $v_B = 0$)

6. Ans (3)

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \Rightarrow v_1 + 3v_2 = 4 \dots (i)$$

$$e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow v_2 - v_1 = 4 \dots (ii)$$

By solving above equations $v_2 = 2 \text{ m/s}$

Momentum of B = $(1.5)(2) = 3 \text{ kg m/s}$

7. Ans (16)

When particle reach at bottom.

From momentum conservation :

$$mv_1 = 4mv_2$$

$$(v_1 = 4v_2)$$

From energy conservation :

$$mgR = \frac{1}{2}mv_1^2 + \frac{1}{2}4mv_2^2$$

$$gR = \frac{1}{2}v_1^2 + 2\left(\frac{v_1}{4}\right)^2$$

$$10 \times 16 = \frac{v_1^2}{2} + \frac{v_1^2}{8} = \frac{5v_1^2}{8}$$

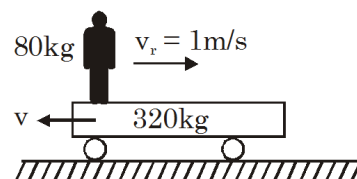
$$v_1^2 = \frac{160}{5} \times 8 = 256$$

$$v_1 = 16 \text{ m/s.}$$

8. Ans (3.20)

$$0 = 80(1-v) - 320v$$

$$v = \frac{8d}{100} = \frac{1}{5} \text{ m/s}$$



$$s_t = \frac{1}{5} \times 4 = \frac{4}{5} \text{ m}$$

$$s_m = \left(1 - \frac{1}{5}\right) \times 4 = \frac{16}{5}$$

$$= 3.2 \text{ m}$$

9. Ans (8.00)

From work energy theorem $W = \Delta KE$

$$W_{\text{gravity}} + W_{\text{air friction}} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$\Rightarrow W_{\text{air friction}} = \frac{1}{2}(4)(14)^2 - (4)(10)(10) = -8J$$

$$\Rightarrow \text{Work done against air friction} = 8J$$

10. Ans (1.00)

$$P = \frac{mgh}{T} = \frac{5 \times 60 \times 10 \times 20}{60} = 1000W$$

$$\left[\because m = 5 \times 60 \text{kg}; h = 20 \text{m and } T = 60 \text{ sec} \right]$$

PART-2 : CHEMISTRY

SECTION-I

2. **Ans (B)**

Given $M_Y = 12 M_X$ (M = atomic mass)

	Mass	Mole	Simplest ratio	Whole no.
Y	7.2	$\frac{7.2}{M_Y} = \frac{7.2}{12 M_X} = \frac{6}{M_X}$	1 = 1	1 × 3 = 3
X	1	$\frac{1}{M_X}$	$\frac{1}{6} = \frac{10}{6} = \frac{5}{3}$	$\frac{5}{3} \times 3 = 5$

E.F. = Y_3X_5

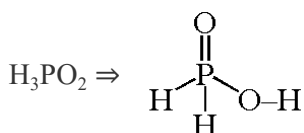
3. **Ans (C)**

Eq. of H_3PO_x = Eq. of KOH

$$\Rightarrow 0.1 \times n = \frac{5.6}{56}$$

$$n = 1$$

So x will be 2



5. **Ans (D)**

$$n_{CaCl_2} = \frac{1.11 \times 10^{-3}}{111} = 10^{-5} = n_{CaCO_3}$$

$$W_{CaCO_3} = 10^{-5} \times 100 = 10^{-3} \text{ gm}$$

$$\text{Hardness} = \frac{10^{-3}}{1000} \times 10^{-6} = 1 \text{ ppm}$$

6. **Ans (B)**

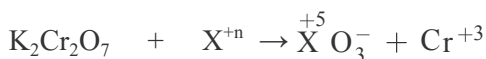
$$Fe_2(C_2O_4)_3 \quad 1 \times 6 = X \cdot 1$$

$$FeC_2O_4 \quad 1 \times 3 = Y \cdot 5$$

$$\frac{6}{3} = \frac{X}{Y \cdot 5}$$

$$\frac{X}{Y} = \frac{10}{1}$$

7. **Ans (A)**



$$n = 6 \quad n = (5 - n)$$

$$6 \times 10^{-3} \times 6 = (5 - n) \times 9 \times 10^{-3}$$

$$\Rightarrow n = 1$$

SECTION-II

4. **Ans (7.50)**

$$\left(\frac{10}{11.35} + \frac{1.5}{11.35} + \frac{20}{11.35} \right) \times 500 = M_f \times 3000$$

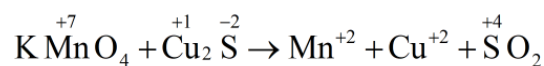
$$M_f = \frac{45}{1135 \times 6}$$

Vol. strength of final solution = $M \times 11.35$

$$= \frac{45}{11.35 \times 6} \times 11.35$$

$$= 7.5 \text{ V}$$

5. **Ans (1.60)**

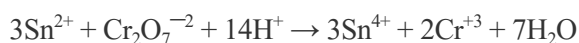


$$nf = 5 \quad nf = 2 + 6 = 8$$

$$5 \times n = 8 \times 1$$

$$n = \frac{8}{5} = 1.6$$

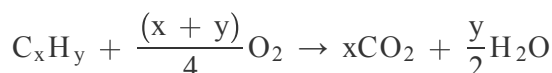
6. **Ans (6.00)**



$$(3 + 1 + 14) - (3 + 2 + 7)$$

$$18 - 12 = 6$$

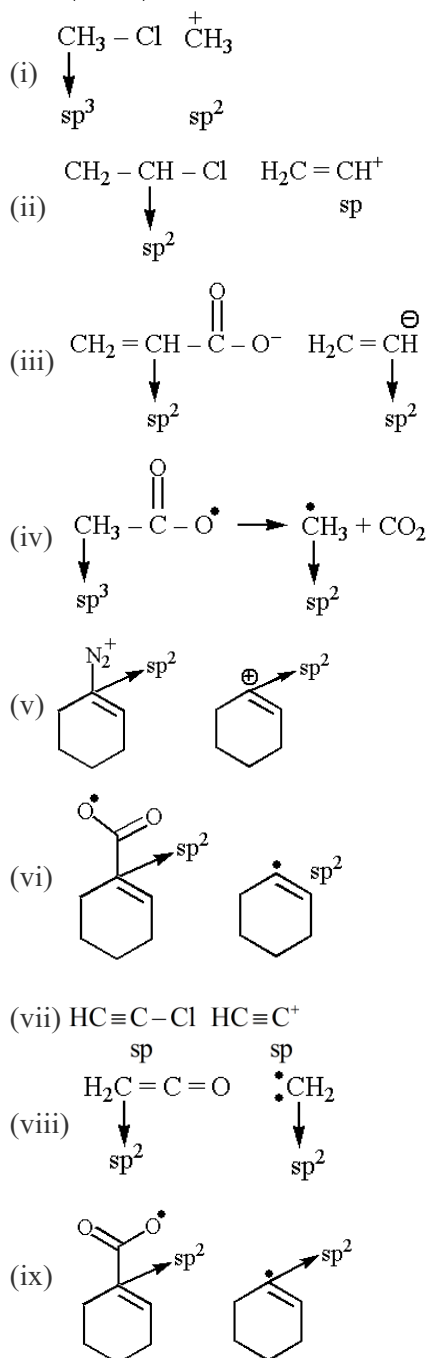
7. **Ans (4.00)**



$$x + \frac{y}{4} = 2x$$

$$\frac{y}{4} = x \Rightarrow \frac{y}{x} = 4.00$$

10. Ans (6.00)



PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)

$$\sqrt{(a-1)^2 + (b-3)^2} = AB$$

$$\sqrt{(4-a)^2 + (7-b)^2} = BC$$

$$\sqrt{(4-1)^2 + (7-3)^2} = AC = 5$$

$\therefore$ minimum value of k is 5

2. Ans (B)

Slope of AB = slope of AC

$$\frac{2}{a-8} = \frac{22}{77} \Rightarrow a = 15$$

3. Ans (A)

$$PB^2 + PC^2 = 2AP^2$$

$$(x+1)^2 + y^2 + (x-2)^2 + y^2 = 2(x-1)^2 + 2y^2$$

$$-3x = -4x + 2$$

$$x = 2$$

4. Ans (B)

$$\frac{1}{k^2} = \frac{2 \left(\frac{1}{k+2} \right) \left(\frac{1}{4k+5} \right)}{\frac{1}{k+2} + \frac{1}{4k+5}} \Rightarrow \frac{1}{k^2} = \frac{2}{5k+7}$$

$$\Rightarrow 2k^2 - 5k - 7 = 0 \Rightarrow k = \frac{7}{2}, -1$$

5. Ans (C)

$$\frac{1}{1.2.3.4} + \frac{1}{2.3.4.5} + \frac{1}{3.4.5.6} + \dots$$

$$T_n = \frac{1}{n(n+1)(n+2)(n+3)}$$

$$= \frac{1}{3} \left[\frac{(n+3) - n}{n(n+1)(n+2)(n+3)} \right]$$

$$= \frac{1}{3} \left[\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)} \right]$$

$$S_n = \frac{1}{3} \left[\frac{1}{1.2.3} - \frac{1}{2.3.4} \right] + \frac{1}{3} \left[\frac{1}{2.3.4} - \frac{1}{3.4.5} \right] + \frac{1}{3} \left[\frac{1}{3.4.5} - \frac{1}{4.5.6} \right] + \dots \infty$$

$$S_n = \frac{1}{3} \left[\frac{1}{1.2.3} \right]$$

$$= \frac{1}{18}$$

6. Ans (B)

$$\text{Degree of LHS} = 3n + 5$$

$$\text{Degree of RHS} = 2$$

$$\Rightarrow 3n + 5 = 2 \Rightarrow n = -1$$

7. Ans (D)

Condition for non-trivial solution

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

$$\Rightarrow 3abc - a^3 - b^3 - c^3 = 0$$

$$\Rightarrow a + b + c = 0, \text{ also product of roots}$$

$$= \frac{c}{a} = (-) \text{ ve}$$

$\therefore$ c and a are of opposite sign.

8. **Ans (B)**

Here

$$D = \begin{vmatrix} 1 & 3 & 2 \\ 1 & 3 & 4 \\ 1 & 3 & 0 \end{vmatrix} = 0$$

$$D_1 = \begin{vmatrix} -2 & 3 & 2 \\ -4 & 3 & 4 \\ 0 & 3 & 0 \end{vmatrix} = 0$$

$$D_2 = \begin{vmatrix} 1 & -2 & 2 \\ 1 & -4 & 4 \\ 1 & 0 & 0 \end{vmatrix} = 0$$

$$D_3 = \begin{vmatrix} 1 & 3 & -2 \\ 1 & 3 & -4 \\ 1 & 3 & 0 \end{vmatrix} = 0$$

So there will be infinitely many solutions.

9. **Ans (D)**

$$(9^y)^2 - (2^x)^2 = 17$$

$$9^y + 2^x = 17 \quad \dots(1)$$

$$9^y - 2^x = 1 \quad \dots(2)$$

$$(1) + (2) \Rightarrow 9^y = 9 \Rightarrow y = 1$$

$$2^x = 8 \Rightarrow x = 3$$

10. **Ans (D)**

$$R_2 \rightarrow R_2 - 2R_1 \text{ \& } R_3 \rightarrow R_3 - 3R_1$$

$$\begin{vmatrix} 1 & x & y \\ 0 & \sin x & \sin y \\ 0 & \cos x & \cos y \end{vmatrix} = \sin(x - y)$$

11. **Ans (D)**

$$D = \begin{vmatrix} a & b & 0 \\ b & c & 0 \\ c & a & 0 \end{vmatrix} \begin{vmatrix} a & b & c \\ b & c & a \\ x & y & z \end{vmatrix} = 0$$

12. **Ans (B)**

$$\log_2 x = \log_7 8$$

$$\Rightarrow \log_2 x = 3 \log_7 2$$

$$\Rightarrow \log_2 x^{\log_2 7} = 3$$

$$\Rightarrow x^{\log_2 7} = 2^3$$

$$\therefore (x^{\log_2 7})^{\log_2 7} = (8)^{\log_2 7} = 7^3$$

$$\therefore \log_3 y = 2 \log_5 3 \Rightarrow \log_3 y^{\log_3 5} = 2$$

$$\Rightarrow y^{\log_3 5} = 3^2 \Rightarrow y^{(\log_3 5)^2} = 3^{2 \log_3 5} = 25$$

$$\text{or } \log_5 z = 2 \log_{11} 5 \Rightarrow z^{\log_5 11} = 5^2$$

$$\Rightarrow z^{(\log_5 11)^2} = 121$$

$$\therefore 7^3 + 5^2 + 11^2 = 489$$

13. **Ans (A)**

$$\alpha + \beta = 3$$

$$\text{also } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$\Rightarrow \alpha\beta = -\frac{7}{4}$$

required quadratic equation is

$$x^2 - x(3) - \frac{7}{4} = 0$$

$$\Rightarrow 4x^2 - 12x - 7 = 0$$

14. **Ans (D)**

$$E = 2c^2 \sin B \cos B + 2b^2 \sin C \cos C$$

$$= 2c^2 \cdot \frac{b}{2R} \cos B + \frac{2b^2 c}{2R} \cos C$$

$$\left(\because \frac{c}{2 \sin C} = \frac{b}{2 \sin B} = R \right)$$

$$= \frac{bc}{R} (c \cos B + b \cos C) = \frac{abc}{R} = 4D = 100$$

15. **Ans (A)**

$$a \sin^2 \frac{B}{2} + b \sin^2 \frac{A}{2} + c$$

$$= a \left(\frac{(s-c)(s-a)}{ac} \right) + \frac{b(s-b)(s-c)}{bc} + c$$

$$= \left(\frac{s-c}{c} \right) (s-a + s-b) + c$$

$$= \left(\frac{s-c}{c} \right) (c) + c = s$$

16. Ans (A)

$$\begin{aligned} \frac{1}{2} + \frac{1}{2 \sin \frac{x}{2}} 2 \sin \frac{x}{2} (\cos x + \cos 2x + \cos 3x + \cos 4x) &= 0 \\ = \frac{1}{2} + \frac{1}{2 \sin \frac{x}{2}} \left(\sin \frac{9x}{2} - \sin \frac{x}{2} \right) &= 0 \\ = \frac{\sin(\frac{9x}{2})}{\sin(\frac{x}{2})} = 0 \Rightarrow x = \frac{2n\pi}{9}, n \neq 9m, m \in \mathbb{I} \end{aligned}$$

17. Ans (B)

$$\begin{aligned} \sin^4 x + \cos^4 x + \sin 2x &= -\alpha \\ 1 - \frac{\sin^2 2x}{2} + \sin 2x &= -\alpha \\ \alpha &= \frac{\sin^2 2x - 2 \sin 2x - 2}{2} \\ \alpha &= \frac{(\sin 2x - 1)^2 - 3}{2} \\ -\frac{3}{2} &\leq \alpha \leq \frac{1}{2} \end{aligned}$$

18. Ans (B)

$$\begin{aligned} a^2 - 2b - 2c &= (\cos A + \cos B + \cos C)^2 \\ -2(\cos A \cos B + \cos B \cos C + \cos C \cos A) &= -2(-\cos A \cos B \cos C) \\ = \cos^2 A + \cos^2 B + \cos^2 C + 2\cos A \cos B \cos C &= 1 \end{aligned}$$

19. Ans (D)

$$\begin{aligned} |\log_{\sqrt{3}} x - 2| - |\log_3 x - 2| &= 2 \\ \Rightarrow |2\log_3 x - 2| - |\log_3 x - 2| &= 2 \\ 2|\log_3 x - 1| - |\log_3 x - 2| &= 2 \\ \text{put } \log_3 x = t & \\ 2|t - 1| - |t - 2| &= 2 \\ \text{Case-1 : } t < 1 & \\ \therefore -2(t - 1) + t - 2 &= 2 \\ t = -2 & \\ \therefore x = \frac{1}{9} & \\ \text{Case-2 : } 1 < t < 2 & \\ \text{On solving } t = 2 \text{ (rejected)} & \\ \text{Case-3 : } t \geq 2 & \\ \text{On solving } t = 2 & \\ \therefore x = 9 & \\ \text{product of solutions} &= 1. \end{aligned}$$

20. Ans (B)

$$\begin{aligned} \text{let } 2^x = t, \text{ let } \alpha < 0 < \beta & \\ \Rightarrow 2^\alpha < 1 < 2^\beta & \\ \Rightarrow \text{given equation has one root less than one} & \\ \text{and other root is greater than 1} & \end{aligned}$$

SECTION-II

1. Ans (1.00)

$$\begin{aligned} \tan x > 0 \text{ \& } \tan x \neq 1 & \\ (\log_{\sqrt{3}} \tan x) \left(\sqrt{\frac{\log_{\sqrt{3}}(\sqrt{3})^3}{\log_{\sqrt{3}} \sqrt{3}}} + \frac{\log_{\sqrt{3}} 3}{\log_{\sqrt{3}} \tan x} \right) &= -1 \\ \text{Let } \log_{\sqrt{3}} \tan x = t < 0 \Rightarrow t \sqrt{3 + \frac{2}{t}} &= -1 \\ 3t^2 + 2t - 1 &= 0 \\ t = -1 \text{ \& } t = \frac{1}{3} \text{ (not possible)} & \\ \therefore \log_{\sqrt{3}} \tan x = -1 \Rightarrow x = \frac{\pi}{6} \text{ only in } [0, \pi] & \end{aligned}$$

2. Ans (3.00)

$$\cos 9x = -1 \quad \cos 7x = 1 \quad x = \pi, 3\pi, 5\pi$$

3. Ans (-4.00)

$$\begin{aligned} (0, 0), \quad (1, 1), \quad (\omega, \omega^2) & \\ x^2 = 0 \quad (x-1)^2 = 0 \quad x^2 + x + 1 = 0 & \\ k = -1, \lambda = 0 \quad k = -3, \lambda = 1 \quad k = 0, \lambda = 1 & \end{aligned}$$

4. Ans (9.00)

$$\Delta = 3 \quad \Delta_{\text{cofactor}} = \Delta^2$$

5. Ans (5.00)

$$\begin{aligned} \frac{a^2 + b^2}{c^2} &= \frac{17}{9} \\ \frac{\cot A + \cot B}{\cot C} &= \frac{\sin C}{\sin A \sin B} \cdot \frac{\sin C}{\cos C} \\ \frac{c^2}{ab} \left(\frac{2ab}{a^2 + b^2 - c^2} \right) &= \frac{2c^2}{a^2 + b^2 - c^2} \\ = \frac{2}{\frac{a^2 + b^2}{c^2} - 1} &= \frac{2}{\frac{17}{9} - 1} = \frac{18}{8} = \frac{9}{4} \end{aligned}$$

6. Ans (2.00)

$$\begin{aligned} \left(\cos A - \frac{1}{\sqrt{2}} \right)^2 + \left(\sin C - \frac{1}{\sqrt{2}} \right)^2 + \cos^2 B &= 0 \\ \text{it is possible when} & \\ \cos A = \frac{1}{\sqrt{2}} = \sin C, \cos B = 0 & \\ A = \frac{\pi}{4} = C, B = \frac{\pi}{2} & \end{aligned}$$

7. Ans (16.50)

$$\begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{vmatrix} = 0$$

$$1(-4k + 6) - k(-12 + 4) + 3(9 - 2k) = 0$$

$$-4k + 6 + 12k - 4k + 27 - 6k = 0$$

$$2k = 33$$

$$k = \frac{33}{2}$$

8. Ans (3.00)

$$|1 - x|^{\log_2(x+1)} > |x - 1|^{\log_2(4-x)}; x \in (-1, 4)$$

$$\text{Case I : } |x - 1| > 1 \Rightarrow x \in (-\infty, 0) \cup (2, \infty)$$

$$\log_2(x + 1) > \log_2(4 - x) \Rightarrow x > \frac{3}{2}$$

$$\therefore x \in (2, \infty)$$

$$\text{Case II : } |x - 1| < 1 \Rightarrow x \in (0, 2)$$

$$\therefore \log_2(4 - x) > \log_2(1 + x)$$

$$\Rightarrow x < \frac{3}{2}$$

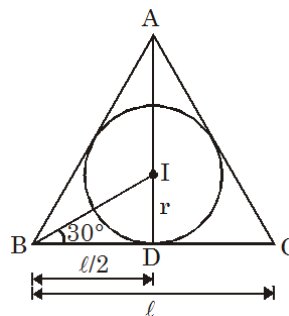
$$\therefore x \in \left(0, \frac{3}{2}\right)$$

but $x = 1$ does not satisfy

$$\therefore \text{final Ans. : } x \in \left(0, \frac{3}{2}\right) \cup (2, 4) - \{1\}$$

$$\therefore \text{only integer value} = 3$$

9. Ans (3.00)



$$\therefore AB = BC = CA = l$$

$$\therefore BD = DC = \frac{BC}{2} = \frac{l}{2}$$

$$\text{In } \triangle BID \Rightarrow \tan 30^\circ = \frac{ID}{BD}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{r}{(l/2)} \Rightarrow r = \frac{l}{2\sqrt{3}}$$

$$\therefore \text{Area of circle} = \pi r^2 = \pi \left(\frac{l}{2\sqrt{3}}\right)^2$$

$$= \frac{\pi l^2}{12} = \frac{\pi l^2}{4\lambda}$$

$$\therefore \lambda = 3$$

10. Ans (5.00)

$$K \cos^2 x - K \cos x + 1 \geq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow K(\cos^2 x - \cos x) + 1 \geq 0 \quad \dots(i)$$

$$\therefore -\frac{1}{4} \leq \cos^2 x - \cos x \leq 2$$

$$\Rightarrow 2K + 1 \geq 0 \Rightarrow K \geq -\frac{1}{2}$$

$$\text{Also } -\frac{K}{4} + 1 \geq 0 \Rightarrow \boxed{K \leq 4}$$

$$\Rightarrow \boxed{-\frac{1}{2} \leq K \leq 4}$$